Finding and Assessing Research Articles with AI

Instructor session plan. 50 minutes. ESS 111 Introduction to Health Careers.

Overview

The existing assignment asks students to find two original research articles that would help them in their chosen profession, describe each in 150 to 350 words of their own writing, and cite them in APA style with a Turnitin score under 20%. This session keeps that assignment intact and adds a worksheet that trains students to find articles, rapidly assess them with AI, check the AI's work, and cite sources with the library cite tool, AI, and EndNote Fusion.

Most students are headed into physical therapist assistant programs or other health professions, with a few other majors. Examples use health research, and the demo topic is physical therapy for low back pain.

Learning objectives

By the end of the session students will be able to

1. Describe in plain language how generative AI produces text and why it can invent research articles that do not exist.
2. Apply the sliding scale of subjectivity to decide how critically to read an AI answer.
3. Have Gemini interview them to build search strings, then find original research articles in Library Search.
4. Tell original research apart from reviews and other article types.
5. Choose between NotebookLM and Gemini for a given task, and use NotebookLM to rapidly assess an article.
6. Verify AI output against the article itself.
7. Cite library articles with the cite tool, web pages with AI, and AI tools themselves in APA style, and know how EndNote Fusion can help with longer papers.

Before class

- ☐ Confirm that **NotebookLM** is turned on for student campus Google accounts, and test uploading a PDF with a student level account.
- ☐ Test whether student Gemini accounts can create chat share links. This affects how students cite their Gemini chat.
- ☐ Check whether any of the library's database licenses restrict uploading full text articles to AI tools. Campus Google Workspace accounts generally keep uploads out of model training, but license terms vary by publisher.
- ☐ Check Library Search on library.byui.edu as students will see it. Confirm the names of the **Peer Reviewed** and **Full Text** limiters and where the **Cite**, **Export**, and **Permalink** tools sit on an article record, and adjust Part 2 and the quick guide if they differ.
- ☐ Prepare the demo. Run Prompt 1 in Gemini, paste one of its strings into Library Search, and choose **one randomized controlled trial** and **one systematic review** to contrast. Download both PDFs and load them into a NotebookLM notebook ahead of time.

- ☐ Look at EndNote Fusion as students will see it. RefWorks was renamed in fall 2026, so check that the library website link, the Export option in Library Search, and the Word and Google Docs add in match the steps in the assignment.
- ☐ Run **Gemini Deep Research** once with the demo prompt (exercise therapy for low back pain). Note how long it takes, whether student accounts have it, and whether its plan and revised searches are easy to show. Save screenshots in case it is slow or unavailable in class.
- ☐ Confirm the open items for the instructor at the end of this plan, especially the rule on AI drafting.
- ☐ Post the assignment as a Word document so students can type into the boxes.

Session at a glance

Clock	Minutes	Segment	Assignment
0:00 to 0:02	2	Welcome and objectives	Header
0:02 to 0:12	10	How generative AI works and three principles	Part 1
0:12 to 0:20	8	Search strings with Gemini, articles with Library Search	Part 2
0:20 to 0:34	14	NotebookLM or Gemini, then rapid assessment	Parts 3 and 4
0:34 to 0:41	7	Citations. Cite tool, web pages, and AI	Part 5
0:41 to 0:45	4	EndNote Fusion demo	Part 6
0:45 to 0:50	5	The paper, AI drafting, and wrap up	Parts 7 and 8

This is tight. The worksheet is due with the paper, so students can finish any part after class. If time runs short, cut the EndNote Fusion demo to two minutes and point students to the setup steps on the resource pages.

Detailed sequence

1. Welcome and objectives

Time 0:00 to 0:02 **Assignment** Header

Instructor

- Introduce yourself and connect the session to the instructor's assignment. They will leave with their two articles found and understood, and a head start on the paper.
- Ask students to open library.byui.edu, NotebookLM, and Gemini in three tabs now, so login problems surface early.
- Start a **Gemini Deep Research** run on the projector before you begin. Prompt it to find three original research articles published since 2020 on exercise therapy for low back pain, and to show how it searched. It takes several minutes, so you will return to it in segment 2.

2. How generative AI works and three principles

Time 0:02 to 0:12 **Assignment** Part 1

The brain comparison. About two minutes

- **A neural network is loosely inspired by the brain.** An LLM is a network of simple artificial neurons joined by billions of weighted connections. Those weights play roughly the role of synapses.
- **Training strengthens useful connections.** Each time the model guesses the next word wrong during training, its weights are nudged. The connections that predict well get stronger, much as practice strengthens synapses.
- **More is not the whole story.** Larger models with more parameters are generally more capable. In the brain, young children have more synapses than adults, and the brain prunes unused ones. Intelligence depends on well organized connections more than a raw count.
- **The key limit.** The model predicts what usually comes next. It has no way to know whether something is true.

The prediction game. About three minutes

- **Round 1. Familiar.** "Twinkle, twinkle, little star, how I wonder…" Nearly everyone agrees. That is a high probability prediction, like AI answering a basic anatomy question.
- **Round 2. Open ended.** "The best treatment for low back pain is…" Answers scatter. When many continuations are plausible, the answer depends on who is predicting. This previews the sliding scale and today's demo topic.
- **Round 3. Familiar is not verified.** "I, Nephi, said unto my father…" Most students will finish it with "I will go and do the things which the Lord hath commanded." Now ask for the exact reference and for the verse that comes right after it. Confidence drops fast. Gemini behaves the same way. It gets the famous line right because it appears everywhere in its training data, but ask for a less familiar neighboring verse and it may produce something fluent and scriptural sounding that is not in the text. That is a hallucination, a fluent guess where the pattern runs thin. Try this in Gemini before class so you know what it does. **Connect it to research.** Ask Gemini for five studies on a topic and some may be invented, with real sounding authors and journals and no actual paper behind them. Sounding right is not the same as being checked against the original.

Not a deterministic program. About one minute

- **Show a simple math program.** A = 3, B = 5, C = A + B. Run it once or a hundred times and C is always 8. That is a deterministic program. The same input always gives the same output.
- **An LLM is different.** Ask Gemini the same question twice and the answers are similar but never exactly the same. Each word is chosen from a range of likely next words, not calculated the way C is.
- **Why it matters.** One AI answer is a sample, not the answer. Two students asking for search terms or article summaries will get different wording, so each still checks against the article itself.

From predicting to problem solving. About two minutes

- **Still predicting, now in a loop.** Everything so far describes a prediction model. Newer reasoning agents still predict the next word, but they are trained to work through a problem in steps and can use tools such as web search. They plan, act, look at the result, and try something else when it does not work.
- **Three generations, one task.** Ask an early prediction model to find articles on low back pain and it writes the most likely looking answer in one pass, invented details included. A chat assistant gives a better answer in one try but does not check itself. A reasoning agent makes a plan, searches, reads what it finds, notices a dead end, changes its search terms, and tries again.

- **Picture a maze.** A prediction model draws the whole path in one guess. An agent walks it, backs up from dead ends, and tries another branch.
- **Return to the Deep Research run you started at the beginning of class.** Show its research plan and scroll through its steps so students can see it revise its searches. Point out the sources it chose.
- **Honest caveat.** The loop makes AI more persistent, not more truthful. An agent can still misread a source or trust a weak one. Students still open every article it lists and read the Methods for themselves.

Three principles. About four minutes

- **The sliding scale of subjectivity.** The more subjective the topic, the more critical you must be. Anatomy and glycolysis sit at the objective end. Contested issues such as the Palestinian situation sit at the subjective end, where AI output is one viewpoint among many. Numbers and citations are a special case that must always be checked.
- **Have AI verify itself, then verify it yourself.** Ask where each claim came from and how confident it is. **Honest caveat.** Self checks catch some errors but not all, and a model will sometimes confidently reconfirm a mistake. The final check is a human reading the article.
- **Let the AI interview you.** When a prompt is vague the AI fills gaps with assumptions. Asking it to interview you first surfaces those assumptions. Students will use this in the very next step.

> **Optional demo prompt for the subjective end**
>
> Give me the strongest perspectives held by different groups on [ISSUE]. For each one, explain who holds it and what evidence they point to. Then tell me what kinds of sources I should read to judge these claims myself.

Students

- Fill in the prediction game (1a) as you go. Questions 1b and 1c can be finished later.

3. Search strings with Gemini, articles with Library Search

Time 0:12 to 0:20 **Assignment** Part 2

Talking points

- **Library Search** is the main search box on library.byui.edu. It is the library's EBSCO Discovery Service (EDS), which searches most of the library's databases at once. Students only need to know it as Library Search.
- **Each tool does one job.** Gemini builds the search strings. Library Search finds the articles. Everything Library Search returns actually exists, which ties straight back to round 3.

Instructor demo

- Run Prompt 1 in Gemini as a future physical therapist assistant interested in low back pain. Answer two or three of its interview questions aloud, then show the search strings it suggests.
- Point out the last line of the prompt. It tells Gemini not to suggest article titles, because invented articles are the round 3 risk.
- Paste one of Gemini's strings into Library Search. Point out the quotation marks, the OR groups in parentheses, and AND between ideas. Then check the **Peer Reviewed** and **Full Text** limiters and note how the result count drops.
- If the results are off, show the fix. Tell Gemini what happened in the same chat and ask for a revised string. Keep this quick. Students only need two articles.

- Show your two prepared results side by side. The systematic review summarizes other studies. The randomized controlled trial has participants, Methods, and Results. Only the trial counts as original research for this assignment.

> **Prompt 1. The search string interview (Gemini)**
>
> I am a first year college student preparing to become a [YOUR PROFESSION, for example physical therapist assistant]. I need to find two original research articles that would help me in that profession. I will search for them in Library Search, my university library's main search box, which searches many scholarly databases at once.
>
> Before you suggest anything, interview me. Ask me one question at a time about the kind of patients or clients I want to work with, the problems I am curious about, and what I already know. Stop after no more than five questions.
>
> Then give me three search strings I can paste straight into Library Search. In each string, put exact phrases in quotation marks, group synonyms with OR inside parentheses, and join two or three different ideas with AND. One idea can be a word like trial OR participants, which helps find original research instead of reviews. Explain each string in one sentence. Do not suggest specific article titles, because you may invent articles that do not exist.

Students

- Run Prompt 1, paste a string into Library Search with the two limiters on, and start choosing their two articles. The Library Search quick guide on the resource pages covers the search tips. Most will need a few more minutes after class to settle on their final two.

4. NotebookLM or Gemini, then rapid assessment

Time 0:20 to 0:34 **Assignment** Parts 3 and 4

When to use which. About three minutes

- Walk through the comparison table below, which also appears in Part 3 and on the resource pages.
- **The short version.** NotebookLM when you have the source. Gemini when you do not.
- NotebookLM is grounded in what you upload and cites the passage behind each answer, so it is much less likely to invent things. It can still misread a study or overstate a finding.
- Have students do the four item matching exercise in 3a quickly, or do it together aloud.

	NotebookLM	Gemini
What it works from	Only the sources you upload, such as your two article PDFs	Its general training plus web searches
Strengths	Answers are tied to your sources, with numbered citations you can click to see the exact passage. Good for summarizing, comparing two articles, and finding the Methods and Results	Knows a little about almost everything. Good for building search strings for Library Search, explaining unfamiliar words like randomized controlled trial, checking APA formatting, and citing web pages
Weaknesses	Knows nothing beyond your sources. Can still misread or oversimplify a study. Its summaries often reuse the article's wording, which raises your Turnitin score	Much more likely to invent facts, statistics, and even whole articles that do not exist. Sounds just as confident when it is wrong

	NotebookLM	Gemini
Use it when	You have the article and want to understand what it says	You do not have a source yet, or you need general help

Instructor demo. About four minutes

- In your prepared notebook, run Prompt 2 on the randomized controlled trial. Click a numbered citation to show the exact passage.
- Run Prompt 2 on the systematic review. NotebookLM should identify it as a review. If it does not, that is a teaching moment.
- Run Prompt 3, the self check, and compare one number with the PDF on screen.

Prompt 2. The rapid assessment (NotebookLM)

Using only the source [ARTICLE TITLE], answer these four questions. Keep each answer to two or three sentences and cite the passage you used.

1. Research type. Is this original research, where the authors collected or analyzed their own data? Or is it a review, a meta analysis, an editorial, or something else? What in the article tells you?

2. What the study did. Who were the participants, how many were there, and what did the researchers do or measure?

3. What the study found. What were the main results?

4. Relevance. How could these findings matter to a future [YOUR PROFESSION]?

If the article does not answer one of these questions, say so instead of guessing.

Prompt 3. The self check

Now check your own work. List every number and factual claim in your last answer. For each one, quote the exact sentence from the article that supports it and tell me which section it comes from, such as Methods or Results. Rate your confidence as high, medium, or low. Flag anything you cannot find in the article and correct any mistakes.

Students. About seven minutes

- Upload both PDFs, run Prompts 2 and 3 for each article, and check one claim from each article against the PDF in 4a. Anyone still searching can use the demo articles to practice and repeat the steps later.

5. Citations. Cite tool, web pages, and AI

Time 0:34 to 0:41 **Assignment** Part 5

Talking points

- **Library articles.** Use the Cite button on the article record in Library Search and choose APA 7th edition. Show the common fixes. Sentence case titles, italics for the journal and volume, and a working DOI.
- **Checking with AI.** Prompt 4 has Gemini check formatting only. It is told not to change facts, because it will sometimes fix a correct detail into a wrong one.

- **Web pages.** Pages from groups like the American Physical Therapy Association have no cite button. Prompt 5 has AI format a citation from details the student supplies. Asked to recall a source from memory, AI invents details.
- **Citing AI.** Students add an AI use statement and APA references for NotebookLM and Gemini. APA guidance on citing AI is still changing, so point them to the Library Citation Guide Videos as the final word.

Prompt 4. Check a library citation (Gemini)

Check this APA 7th edition reference for a journal article and tell me about any formatting errors. Look at author names and initials, the year, sentence case in the article title, title case and italics for the journal name, italics for the volume, the issue and page numbers, and the DOI written as https://doi.org/ followed by the number.

Do not change any facts such as names, dates, titles, or numbers. If a fact looks wrong, tell me to check it against the article instead of fixing it yourself.

Here is the reference. [PASTE THE REFERENCE FROM THE LIBRARY CITE TOOL]

Prompt 5. Cite a web page (Gemini or NotebookLM)

Create an APA 7th edition reference list entry and a matching in text citation for the web page below. Use only the details I give you. Do not invent an author, date, or web address. If something APA requires is missing, tell me what it is instead of guessing.

Source details. Page title [ ]. Author or organization [ ]. Publication or last updated date, or write none [ ]. Website name [ ]. Web address [ ].

Students

- Copy the APA reference for each article into 5a. The web page practice in 5b can be done after class.

6. EndNote Fusion demo

Time 0:41 to 0:45 **Assignment** Part 6

Instructor demo

- Frame it honestly. With two sources this paper does not need a citation manager, but the longer papers in their programs will.
- Show EndNote Fusion from the library website and mention that it was called RefWorks until this fall, so older guides and some menus still use that name.
- Export your demo trial from Library Search to EndNote Fusion, and drag in the review PDF to show that it reads the details.
- Open Word, show the EndNote Fusion add in (formerly RefWorks Citation Manager), insert an in text citation, and let it build the bibliography. Mention that it also works in Google Docs.
- Mention that the older Write N Cite plugin no longer works, since students may find it in older guides.

Students

- Create an account and save one article if time allows. Installing the add in can wait until they write.

7. The paper, AI drafting, and wrap up

Time 0:45 to 0:50 **Assignment** Parts 7 and 8

The paper

- Walk through the instructor's requirements in Part 7. Two articles, 150 to 350 words each, no quotations, in text citations, an APA bibliography, and Turnitin under 20%.
- Show Prompt 6. Students may have NotebookLM draft, but they must substantially revise. Explain why. AI summaries reuse the article's phrases, which raises the similarity score, and Turnitin also flags AI generated writing separately.
- Point out the revision checklist and the AI use statement.

> **Prompt 6. Draft a paragraph (NotebookLM)**
>
> I need to write one or two paragraphs, 150 to 350 words total, about the source [ARTICLE TITLE] for a class assignment. Before you write anything, interview me one question at a time about my future profession and why this article interests me. Ask no more than four questions.
>
> Then write a draft that explains why I chose the article, what the study did and found, and how the information will help me in my profession. Use plain language. Do not use any quotations, and do not copy phrases from the article. Paraphrase everything.

Key takeaways

- AI predicts. It does not look things up, so never trust an article it names until you find it in Library Search.
- NotebookLM when you have the source. Gemini when you do not.
- Let the AI interview you before it works for you.
- Ask the AI to verify itself, then verify it yourself against the article.
- A draft from AI is a starting point. The paper has to be yours.

Students

- Finish the worksheet, write the paper, and submit both together by the instructor's due date.

Suggested grading

A draft combined rubric for the instructor to adjust to their point scale. Thirty points total, with the paper carrying most of the weight.

Section	Points	What earns full credit
Paper. Article 1 write up	6	150 to 350 words that explain why the student chose it, what the study did and found, and how it will help them. Original research. Own words with no quotations
Paper. Article 2 write up	6	Same as article 1
Paper. APA citations	4	In text citation after each write up and a correct APA bibliography of both articles
Paper. AI use statement and Turnitin	4	An honest AI use statement with APA references for the AI tools. Turnitin similarity under 20%

Section	Points	What earns full credit
Worksheet. Parts 1 to 3	3	Prediction game, sliding scale, articles recorded, and the tool choice exercise completed with specific answers
Worksheet. Parts 4 to 6	4	Two AI claims checked against the articles. APA references checked. EndNote Fusion account created
Worksheet. Part 8 reflection	3	Shows genuine thinking about AI in their profession and appropriate use in general

Troubleshooting

Problem	What to try
Deep Research is slow, unavailable, or still running	Show the screenshots from your practice run and come back to the live run at the end of class if it finishes.
NotebookLM is not available on a student account	Have the student use Gemini with the PDF uploaded and the same prompts, and stress checking every claim, since Gemini is less tightly grounded.
A PDF will not upload or reads as blank	Some PDFs are scanned images. Download the PDF again, or copy the article text into NotebookLM as a pasted text source.
NotebookLM calls a review original research	That is the lesson. Have the student check the Methods section and the original research checklist.
The cite tool leaves out the DOI or gets the capitalization wrong	Common. Have the student fix it using the checklist in Part 5 or Prompt 4.
Library Search returns thousands of results or almost none	Have the student tell Gemini what happened in the same chat and ask for a narrower or broader string. The quick guide on the resource pages has the same tips.
An article asks the student to pay	The student likely landed on the publisher site from Google. Open the article from Library Search while signed in, with the Full Text limiter on.
Student cannot find EndNote Fusion	It may still be listed as RefWorks on the library website.

Open items for the instructor

- **AI drafting and Turnitin.** The assignment lets students have AI draft if they substantially revise. Turnitin's AI writing indicator is separate from the 20% similarity limit. Decide whether a high AI indicator affects the grade, and say so to students.
- **Point scale.** Adjust the suggested rubric to match his existing points for this assignment.
- **Due date.** The worksheet is due with the paper. Confirm the date.
- **Library Citation Guide Videos.** Add the link from his original assignment to the course page or to Part 5.

Sources consulted

- Google. NotebookLM Help. support.google.com/notebooklm
- University of Wisconsin Madison Libraries. Citing Generative AI (summary of APA 2025 guidance). researchguides.library.wisc.edu/GenerativeAI/Citing
- EndNote Fusion. Introducing EndNote Fusion. fusion-endnote.zendesk.com
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